MIL-M-22525B(SHIPS)
15 May 1972
SUPERSEDING
MIL-L-22525A(SHIPS)
31 May 1967
(See 6.5)

MILITARY SPECIFICATION

MINE FIELD MARKER, BF MK 3 MOD 0, NM (HIGH INTENSITY FLASHING BEACON LIGHT)

1. SCOPE

1.1 This specification covers a nonmagnetic high intensity flashing light primarily intended for use on minefield marker (reference) buoys as an aid to navigation. It is designed to be as compact as possible and yet to be rugged enough to operate under adverse environmental conditions. The light consists of two basic components, an electronic flash assembly and a battery pack which are interconnected by a power cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-18 - Batteries, Dry.
MIL-P-116 - Preservation, Methods of.
MIL-I-17214 - Indicator, Permeability; Low-MU (GO-NO-GO).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories And Repair Parts; Packaging and Packing of.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

MILITARY

403-4302,211 (7300) Light Assembly, Minefield Marker, Flashing.
403-4302,211 (7301) Schematic Drawing, Light, Minefield Marker, Flashing.
403-4302,211 (7302) Battery Container Assy., Light, Minefield Marker, Flashing.
403-4302,211 (7303) Battery Tube, Minefield Marker.
403-4302,211 (7304) Contact, Minefield Marker.
403-4302,211 (7305) Contact, Minefield Marker.
403-4302,211 (7306) Spacer Tube, Light, Minefield Marker, Flashing.
403-4302,211 (7307) Pad, Minefield Marker.
403-4302,211 (7308) Cap Assy., Light, Minefield Marker, Flashing.
403-4302,211 (7309) Cap, Minefield Marker.
403-4302,211 (7310) Handle, Minefield Marker.
403-4302,211 (7311) Pipe, Minefield Marker.
403-4302,211 (7312) Cable Assy., Battery Container, Light, Minefield Marker, Flashing.
403-4302,211 (7314) Rod, Minefield Marker.
403-4302,211 (7315) Electronic Assy., Minefield Marker, Flashing.
403-4302,211 (7316) Printed Circuit Board Master, Minefield Marker.
403-4302,211 (7317) Designations Markings, Minefield Marker, Flashing.
403-4302,211 (7318) P/C Board Machining, Minefield Marker, Flashing.
403-4302,211 (7319) Light Tube, Minefield Marker.
403-4302,211 (7320) Bracket, Minefield Marker.
403-4302,211 (7321) Clip, Minefield Marker.
403-4302,211 (7322) Guard, Light, Minefield Marker, Flashing.
403-4302,211 (7323) Washer, Insulating, Minefield Marker.
403-4302,211 (7324) Tubing, Mask, Minefield Marker.
403-4302,211 (7325) Cable Assy., Electronic Container, Light, Minefield Marker, Flashing.

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MILITARY (Cont'd)

403-4302,211 (7326) Specifications Converter-XMFR, Minefield Marker.
403-4302,211 (7327) Clamp Assy., Light, Minefield Marker, Flashing.
403-4302,211 (7328) Latch Assy., Minefield Marker.
403-4302,211 (7329) Latch, Minefield Marker.
403-4302,211 (7330) Strap, Latch, Minefield Marker.
403-4302,211 (7331) Lockplate, Light, Minefield Marker, Flashing.
403-4302,211 (7332) Dee Ring Minefield Marker.
403-4302,211 (7334) Clamp Sub-Assy., Minefield Marker.
403-4302,211 (7335) L. H. Clamp Segment, Minefield Marker.
403-4302,211 (7336) R. H. Clamp Segment, Minefield Marker.
403-4302,211 (7337) Light Tube Assy., Light, Minefield Marker, Flashing.
403-4302,211 (7338) Tube Socket, Minefield Marker.

PUBLICATIONS

NAVSHIPS

0981-012-1000 - High Intensity Flashing Beacon Light, Minefield Marker, BF MK 3 MOD 0 - Operation.
0981-012-2000 - High Intensity Flashing Beacon Light, Minefield Marker, BF MK 3 MOD 0 - Maintenance.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Sample for first article inspection. The first five lights produced under the contract shall be tested as specified in 4.2 (see 6.2).

3.2 Construction. The flashing light shall be constructed in accordance with the drawings listed in 2.1.

3.2.1 Batteries. Batteries shall be furnished with the lights but shall be packaged separately (see section 5).

3.2.2 Materials. Materials and parts used to construct the light shall be those specified on the drawings listed in 2.1 or direct equivalents thereto. Except for those applications where the function of the component requires the use of magnetic material, all components shall have a magnetic permeability not exceeding 2.0 after fabrication, when tested with a permeability indicator in accordance with 4.5.7.

3.2.3 Interchangeability. Every part of each light shall be completely interchangeable, mechanically and electrically, with the corresponding part of every other light.

3.3 Performance. Lights shall be capable of meeting the performance requirements specified in 3.3.1 through 3.3.6 to insure that the electronic and other components affecting performance and watertight integrity of the lights and the assembly of these components are as specified on the drawings (see 2.1).

3.3.1 Flash rate. The basic initial flash rate shall be 110 +5 flashes per minute. The manual selector shall provide an alternate initial flash rate of 60 ±15 flashes per minute with a lower flash intensity. The manual selector shall be designed in such a way that the mode cannot be inadvertently changed.

3.3.2 Light sensitive switch. The light sensitive switch shall turn the light on automatically at dusk or when the illumination level drops below 1.5 +0.5 foot candles. It shall also turn the light off automatically at dawn, or when the light level reaches 3 ±1 foot candles.

3.3.3 Operating life. In a temperature environment of not more than 85°F., the light shall operate without battery replacement for at least nine consecutive days (12 hours on, 12 hours off) at a minimum of 90 flashes per minute in the High Flash Rate Mode. It shall operate in the same environment without battery replacement for at least twelve consecutive days (12 hours on, 12 hours off) at a minimum of 30 flashes per minute in the Alternate Flash Rate Mode.

3.3.4 Operating temperature. The light shall start at all temperatures from 32° to 110°F. and shall be adjustable to the flash rates specified in 3.3.1.

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3.3.5 Submergence. The light shall be watertight to external water pressures of 10 pounds per square inch (psi).

3.3.6 Visibility. The light shall have an initial visibility range at the high flash rate of at least 8,000 yards when there is an atmospheric light transmission of 70 percent per mile and background brightness of 0.0001 foot lambert. At the end of the life span specified in 3.3.3, the light shall have a visibility range at the high flash rate of at least 7,200 yards under the same visibility conditions.

3.4 Manuals. When required (see 6.1), manuals (manuscript only), including reproducible art work, shall be prepared in accordance with 3.4 through 3.4.2 containing information necessary for operation and maintenance of the lights. Parts identification shall be included and shall contain brief descriptive information for basic and repair parts to facilitate identification of the parts in the assembly, and to permit ordering of replacement parts. Graphics used to supplement the descriptive matter contained in the text for parts identification shall consist of photographs or line drawings or both. Color shall not be employed. The manuscript should be identical in organization, format and content to NAVSHIPS 0981-012-1000 and 0981-012-2000 wherever the new manuals agree with the current manuals.

3.4.1 Preliminary section. At least 45 days prior to beginning the final manuscript, the contractor shall forward two copies of the manuscript in preliminary form, together with two copies of each photograph or line drawing required to illustrate the text, to the Naval Ship Engineering Center, SEC 6127, Hyattsville, Maryland 20782 via the cognizant Defense Contract Administration Service (DCAS) and the procuring activity for review.

3.4.2 Final section. Two copies of the final section, together with two reproducible copies of each photograph or line drawing, shall be forwarded via the cognizant DCAS and procuring activity to the Naval Ship Engineering Center, SEC 6127, Hyattsville, Maryland 20782 for reproduction and distribution.

3.5 Identification marking. The electronic assembly case and the battery container shall each be clearly labeled, using subsurface lettering with the following identifying data:

(a) "Minefield marker, BF MK 3 MOD 0, NM (High Intensity Flashing Beacon Light)."
(b) Federal stock number.
(c) Manufacturer's name.
(d) Contract number.
(e) Month and year of manufacture.

3.6 Workmanship. The quality of the workmanship used in the construction of the light shall be such that the intent of the specific directions given in the drawings listed in 2.1 may be achieved.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

4.1.2 Records. Records of all tests shall be kept complete and available to the Government.

4.2 First article inspection. First article inspection shall consist of the tests specified in 4.5. Lights which have successfully passed the first article tests may be submitted for tests as production equipment.

4.3 Sampling for quality conformance tests.

4.3.1 Lot. For purposes of sampling, a lot shall consist of all lights offered for delivery at one time.

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Extract from its appearance or impair its serviceability

4.3.2 Sampling for production tests. From each lot, a random sample of lights shall be selected as shown in table I for the tests specified in 4.4.

Table I - Sampling for production tests.

Number of lights in lot	Number of lights in sample
2 to 8	2
9 to 15	3
16 to 25	5
26 to 40	7
41 to 65	10
66 to 110	15
111 to 180	25
181 to 300	35
301 to 500	50
501 to 800	75
801 to 1300	110

4.4 Production tests. Each light selected in accordance with 4.3.2 shall be subjected to the submergence, flash rate and battery current drain tests specified in 4.5.2, 4.5.3, and 4.5.5. Any light in the sample containing one or more defects shall be rejected and the lot represented by the sample shall be subject to rejection.

4.5 Tests.

4.5.1 Operating test. Starting of the first article lights at 32°F. and 110°F. operating temperatures in each of the flash rates shall be demonstrated as specified in 3.3.4.

4.5.2 Submergence test. The entire light assembly (electronic flash assembly, battery pack and interconnecting power cable) shall be submerged in water at a pressure of 10 psi for a period of 30 minutes. After removal from the water the electronic flash assembly and the battery container shall be disassembled and examined for leakage. There shall be no water leakage past any of the seals or in the power cable connectors as a result of this test.

4.5.3 Flash rate test. The device shall be placed in a darkened room, and the flash rate in flashes per minute shall be determined by averaging the count of the flashes during a 3 minute period. The high flash rate shall have an average of 110 ±5 flashes per minute. The flash rate jumper bar shall then be moved to the low flash rate position and the low flash rate shall be determined in the same manner. The low flash rate shall have an average value of 60 ± 15 flashes per minute.

4.5.4 Flash energy test. The housing shall be removed from the electronics section so that the voltage across the 2 microfarad capacitor C8, shown on Drawing 403-4302,211(7301) can be monitored during operation of the light. The input lead of an oscilloscope, Tetronix Model 516 or equivalent, shall be connected across capacitor C8 while the light is flashing. For the high flash rate and the low flash rate, the voltage wave form observed on the oscilloscope display shall reach the maximum level of 620 ± 50 volts prior to each flash discharge and a minimum level of 20 ± 20 volts after each discharge.

4.5.5 Battery current drain test. The battery current drain shall be measured using a current probe, Tetronix Model P6016, a current probe amplifier, Tetronix Type 131, and a Tetronix Model 516 Oscilloscope or equivalents. The average current drain shall not exceed 400 milliamps in the High Flash Rate Mode and 300 milliamps in the low flash rate mode.

4.5.6 Visibility and life test. The first article lights shall be tested to determine that the visibility requirement of 3.3.6 can be met. This test shall be conducted in a temperature environment of not more than 85°F. over a nine day period. The light shall operate without battery replacement in the High Flash Rate Mode for nine consecutive days (12 hours on, 12 hours off). The flash rate shall not drop to lower than 90 flashes per minute (see 3.3.3).

4.5.7 Permeability test. All applicable components of the light assembly shall be checked with a permeability indicator conforming to MIL-I-17214 to determine that they meet the magnetic permeability requirement specified in 3.2.2.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with Section 5 of this document.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation and packaging. Lights shall be packaged by level A or C, as specified (see 6.1). Each light shall be packaged separately. The batteries required for each light shall be packaged separately by level A or C, as specified and then shall be placed in the package with the lights.

5.1.1 Level A. Unit packaging for the lights shall be in accordance with method III of MIL-P-116. The batteries for each light shall be packaged separately in accordance with level A of MIL-B-18 and shall then be placed in the package with the light.

5.1.1.1 Level C. Unit packaging for each light shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage. This level may conform to the supplier's commercial practice. Batteries for each light shall be packaged separately in a manner which will provide the same degree of protection as specified above for the light and shall then be placed in the package with the light.

5.2 Packing. Lights and batteries shall be packed by level A, B, or C as specified (see 6.1) in accordance with MIL-E-17555.

5.2.1 Marking. Marking shall be as specified in 5.2.2 and 5.2.3. The method and size of the markings shall be in accordance with MIL-STD-129.

5.2.2 Interior packages. The marking information on the interior packages shall be in accordance with the contractor's commercial practice. The information on the package for the light assembly shall include nomenclature, federal stock number, contract or order number, contractor's name, protection level and date.

5.2.3 Exterior shipping containers. The marking information on the exterior shipping containers shall be in accordance with the contractor's commercial practice and shall include the shipping destination in addition to the information listed in 5.2.2.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether manuals are required (see 3.4).
(c) Quality assurance requirements (see 4.1.1).
(d) Level of packaging required for lights (see 5.1).
(e) Level of packing required for lights and batteries (see 5.2).

6.2 Management control system document. The following management control system document should be included on DD form 1660:

(a) MIL-I-45208 (see 4.1.1).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 THE MARGINS OF THIS SPECIFICATION HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATION, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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